Analyzing Aggie Buses and Their Time Accuracy

TEXAS A&M UNIVERSITY
College of Arts & Sciences

Emie McCoy
Department of Geography
Texas A&M University

Abstract

Texas A&M just recently implemented a new bus system with routes designed to be more centralized with the hub being around Simpson Drill Field. To assess the effectiveness of this new system, the estimated time of arrival was compared to actual time stamps recorded by group members. The comparison revealed a sizable error in accuracy. Nearly 60% of buses were not on time according to the schedule. Looking at this data has highlighted some of the ongoing challenges in bus accuracy and the need for improvement so students can have a reliable bus system.

Introduction

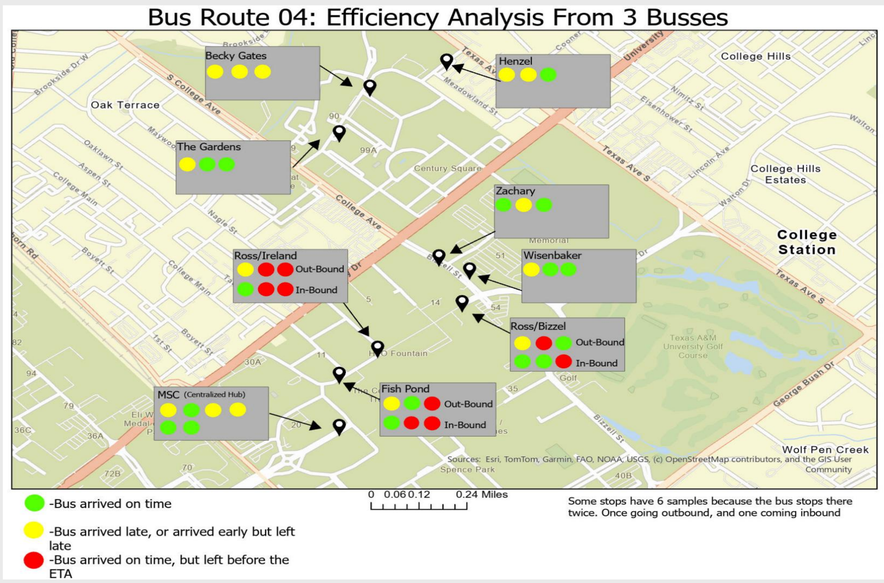
Texas A&M Transportation has recently completely reconstructed their bus system to create a more centralized system. As of August 2025, all buses now make a stop in central campus at the Memorial Student Center. There have however, been many complaints from students about the efficiency of this new bus system, and inaccurate bus ETA's leading to delays and inconveniences.

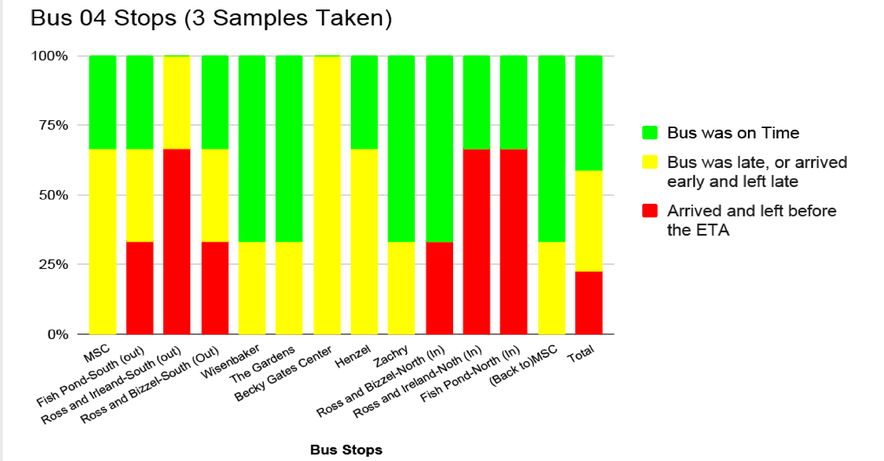
Methods

Looking at the time sheet provided by Aggie Transportation we wanted to view and record two main points. A buses capacity from stop to stop (out of 100%), and the ETA of each stop. The actual time that a bus stopped was document to compare it to the original ETA to see exactly what the exact time difference was and see where the problem spots in/around campus are, and maybe some factors as to why these are problem spots.

- **Green =** bus arrived on time
- **Yellow =** bus arrived later than the ETA, or arrived early and left later than ETA
- **Red =** bus arrived and left before ETA

Results





When looking at the data collected from the 7 bus routes, only about 40% of the buses were on time when compared to the time sheet that is on the Aggie Transportation website. When specifically looking at bus route 04, only 41% of buses were on time making the results almost identical to that of the other 6 routes combined. Having less than 50% of buses not accurately making it to the designated stops on time is troubling and inconvenient for those who rely on it. It was also found that a lot of bus drivers do not accurately track capacity which can also cause some inconveniences.

Discussion

- Which of the 7 routes analyzed has the most issues, and what may be causing these issues to be reoccurring?
- Drivers are not accurately tracking bus capacity, and this can also be troubling because if it is believed to be full people may just decide to Just wait for the next one that is 15-20 minutes away, putting students behind schedule.

Conclusion

The new bus network is a more centralized for every route, but the time accuracy proves to be inaccurate and inconsistent with buses arriving later than the ETA or leaving before the ETA. From the 7 routes analyzed around 60% of all buses were not where they were supposed to be according to the time sheet from Aggie Transportation. It was also observed that bus capacity was not accurately taken. To improve this system to make it more reliable for students, there needs to be an improvement in real-time tracking to help limit the number of delays/early departures. Once these improvements are made it will become an extremely reliable and efficient bus system that students can depend on.

References

Texas A&M University. RouteMap. Aggie Spirit. https://aggiespirit.ts.tamu.edu/RouteMap